

MAG: A Novel Approach for Effective Anomaly Detection in Spacecraft Telemetry Data

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Abstract—Anomaly detection is a crucial matter to ensure the spacecraft stability. During the spacecraft operation, sensors and controllers generate a large volume of multidimensional time series telemetry data with long periodicity, and one key point to detect the anomaly inside the spacecraft timely and precisely is to extract essential features from the sheer amount of telemetry data. However, great challenges exist owing to the complex coupling relationships and the temporal characteristics inside the telemetry data. To address this issue, we propose a novel approach called maximum information coefficient attention graph network (MAG). The basic frame is a graph neural network, which utilizes embedding vectors to describe the intrinsic properties of each dimension, correlation analysis to investigate long-term dependencies, an attention mechanism to determine short-term interactions among dimensions, and long short term memory (LSTM) to extract temporal features. The fusion of these modules through a graph neural network results in the construction of the MAG model, allowing for a comprehensive analysis of complex variable relationships and temporal characteristics leading to successful detection of various types of anomalies. Since telemetry data has heterogeneous characteristics, we adapt the loss function and design an unsupervised anomaly scoring method suitable for MAG. To verify the effectiveness of the proposed algorithm, we conducted experiments using two publicly and two new available spacecraft telemetry datasets, and the results demonstrate that our algorithm is more efficient and accurate in detecting spacecraft data anomalies than several other state-of-the-art methods.

Index Terms—Anomaly detection, anomaly score, graph neural network (GNN), multivariate time series, spacecraft telemetry data.

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I. INTRODUCTION

ENSURING the safety of spacecraft systems holds immense importance due to their complexity and magnitude [1]. In particular, even a minor malfunction can result in catastrophic destruction of the spacecraft. Hence, early detection of anomalous behavior in the spacecraft system during the development of a fault becomes critical to prevent it from exacerbating into a catastrophic failure.

Due to the complexity of dimension interrelationships and high-dimensional nature of spacecraft telemetry data, traditional modeling techniques encounter challenges in anomaly detection. As a result, data-driven methods that rely on telemetry data have gained significant attraction [2]. Given that anomalies are usually rare and few in number, unsupervised learning solely using normal telemetry data to uncover inherent patterns holds greater value. The key to anomaly detection using only normal telemetry data lies in extracting essential characteristics of spacecraft normal operation states from multidimensional time series data. However, due to the complex interrelationships and temporal characteristics of telemetry data, this presents a significant challenge.

In response to this challenge, numerous scholars have proposed various classic unsupervised anomaly detection models that have demonstrated excellent performance in various fields. These include local outlier factor (LOF) [3], one-class support vector machine (OC-SVM) [4], and support vector data description (SVDD) [5], etc. [6]. Moreover, conventional machine learning methods are insufficient to address the challenge of handling high-dimensional and large-scale data. As artificial intelligence has developed, deep learning has increasingly addressed this issue [7]. For instance, Su et al. [8] and Song et al. [9] proposed anomaly detection methods based on the GRU-VAE and ST-GAN, respectively. These methods exhibited solid performance on publicly available datasets.

However, those previously mentioned are primarily designed for Euclidean domain data with translational invariance, while the telemetry data are better mapped to non-Euclidean domain data, such as graph structures [10]. For this reason, graph neural networks are well suited for telemetry data, where graph nodes represent the features of variables within the detection window, and graph edges describe the correlations between dimensions. Indeed, Deng et al. [11] proposed a bias score-based anomaly detection algorithm, and Xie et al. [12] suggested an anomaly

detection method that combines wavelet transform and graph neural networks, both with commendable performance on industrial datasets.

Nonetheless, these methods above are not entirely suitable for our problem. In the case of complex, multidimensional, and time-sensitive telemetry data, it is vital to accurately capture the intrinsic and temporal characteristics of each variable within each detection window, while extensively examining their intricate nonlinear interdependencies. Furthermore, telemetry data characterize long-period properties, making methods that rely solely on short-term windows for extracting correlation capabilities insufficient. Moreover, telemetry data obtained from spacecraft systems contain two heterogeneous data types: analog and state variables (analog variables are continuous and state variables are binary), which render a single loss function inadequate.

Based on the above analysis, this article proposes a novel graph neural network framework. The inherent and temporal characteristics of each variable are represented as nodes, while the long-term association analysis and attention within the short-term window are depicted as edges. Ultimately, the entirety of the graph structure undergoes updates to finalize the anomaly detection process. The main contributions and advantages of our work are as follows.

- 1) This article aims to represent the effective features in the telemetry data detection window from three angles: by utilizing embedding vectors to describe the properties of each dimension, demonstrating the relationships between dimensions using graph structures, extracting temporal features using LSTM, and fusing the various components via a graph neural network.
- 2) To account for telemetry data's long periodicity, this article uses maximum information coefficient (MIC) to investigate the long-term relationships of variables, an attention mechanism to capture the short-term associations of windows, and ultimately combines both methods to construct edges in the graph that represent the associations.
- 3) This article tackles the heterogeneity of telemetry data along with the feature of edge fusion construction and puts forward a novel loss function. Additionally, a threshold score appropriate for the network structure is devised.

The rest of this article is organized as follows. Section II describes the problem of spacecraft system telemetry data anomaly detection. Section III presents the overall framework of our approach to anomaly detection. Section IV describes the proposed maximum information coefficient attention graph network (MAG) algorithm in detail. The experimental results are reported in Section V. Finally, Section VI concludes the article.

II. PROBLEM DESCRIPTION

The types of spacecraft telemetry data can be classified into analog variables (e.g., piezoelectric voltages, acceleration quantities) and state variables (e.g., module control commands). Anomalies are typically indicated as either outliers or subseries of multiple telemetry variables in the telemetry data, encompassing point anomalies, shapelet anomalies, trend anomalies,

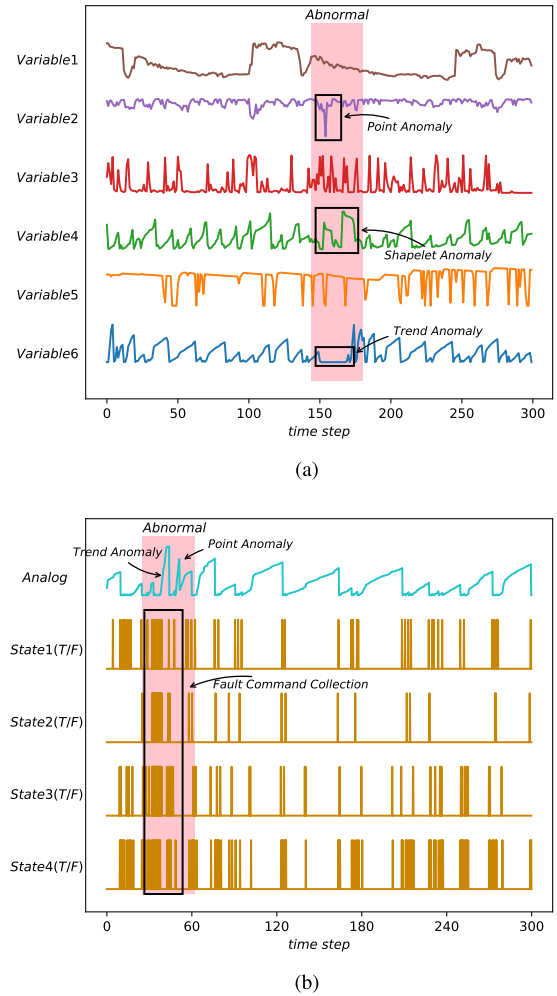


Fig. 1. Illustrative examples of anomalies in spacecraft telemetry data. (a) Telemetry analog variables. (b) Telemetry of analog variables and state variables.

contextual anomalies, and multiset anomalies. In Fig. 1(a), there are inconsistencies in the periods among the variables. Variable 1, in particular, exhibits a longer period that may exceed the window length used for detection. This feature also highlights the inadequacy of analyzing correlation solely within the length of the detected time window. Thus, it is imperative to conduct association analysis of the entire sequence. The anomaly consisting of analog variables and state variables is shown in Fig. 1(b). State 1 to State 4 are commands sent or received by the console. During the anomaly interval, trend anomalies and point anomalies were observed for the analog variables, which were obtained by ground testers actively entering incorrect module commands.

The method for achieving real-time anomaly detection involves utilizing sliding window data to make real-time predictions for multivariate time series, and then determining whether the prediction results deviate excessively from the observed values. If the deviation is too large, the prediction is deemed to be anomalous. Defining $X = \{x^{(1)}, x^{(2)}, \dots, x^{(T)}\}$ as the original time series with time span T , $\hat{x}^{(T+1)}$ as the forecast

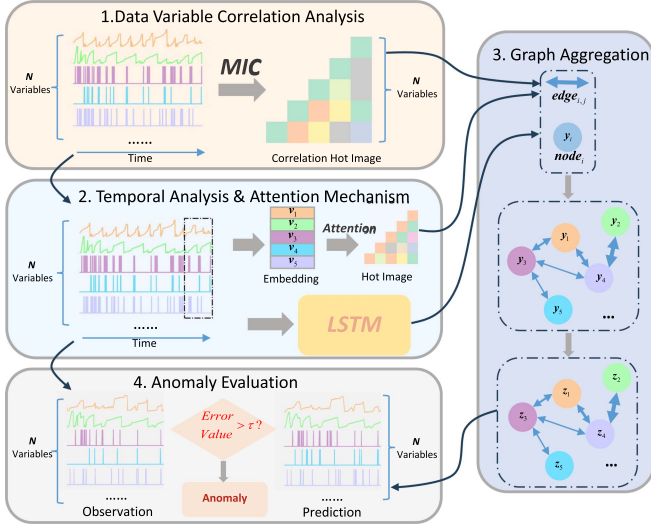


Fig. 2. Overview of the proposed framework.

value of this time series at the moment $T + 1$ after the model, and $x^{(T+1)}$ as the true telemetry value at the moment $T + 1$, then whether it is anomalous can be expressed by

$$\|x^{(T+1)} - \hat{x}^{(T+1)}\| > \tau \quad (1)$$

where $\|\bullet\|$ denotes the rule to calculate the deviation between $\hat{x}^{(T+1)}$ and $x^{(T+1)}$, τ is a predefined anomaly threshold. The larger the deviation between $\hat{x}^{(T+1)}$ and $x^{(T+1)}$, the higher the anomaly probability of $x^{(T+1)}$. As shown by the anomalies of the data in Fig. 1, the key to problem is to find the temporal and variable correlations of the data. We propose the model MAG to solve this problem and predict $\hat{x}^{(T+1)}$, and propose a new formula to find the threshold τ .

III. PROPOSED FRAMEWORK

The proposed MAG framework aims to capture the correlation and temporal characteristics among telemetry data variables, integrating them into a graph. Subsequently, the framework utilizes the aggregated updates from the graph to predict future variable outcomes and detect anomalies by comparing them with the observed values.

The overall structure of the proposed framework is shown in Fig. 2. It consists of the following four main parts.

- 1) *Variable Correlation Analysis*: The telemetry data obtained from the spacecraft system are analyzed using the MIC algorithm to obtain the correlation coefficient matrix, reflecting the correlations between variables during normal working conditions.
- 2) *Temporal Analysis and Attention Mechanism*: The data are subjected to sliding window segmentation. LSTM is employed to extract temporal features from the windowed time series, while employing embedding vectors to capture the inherent attributes of each dimension. The attention mechanism is utilized to obtain the attention coefficient matrix.

- 3) *Graph Aggregation*: The correlation coefficient matrix and attention coefficient matrix are integrated as edges, while the time feature nodes and embedding vectors are incorporated as node features to construct a graph. Subsequently, the entire network is aggregated and updated through iterative processes.
- 4) *Anomaly Evaluation*: Ultimately, the MAG network predicts the next variable value and compares it with the observed value, generating an error score. This score is then used to determine the presence of abnormalities.

IV. DETAILS OF THE PROPOSED METHOD

A. Data Preprocessing

Partitioning the multivariate telemetry data is imperative for real-time calculations, which requires the establishment of real-time calculation detection windows. First, split multivariate telemetry data $\Phi \in R^{T \times N}$. The data have a total length of T , consist of N variables, a set of training data is denoted $\Phi_{\text{train}} \subseteq R^{T_1 \times N}$, and a set of test data is denoted $\Phi_{\text{test}} \subseteq R^{T_2 \times N}$. Note that all the data points in the training data set must be normal.

Next, the training dataset $\Phi_{\text{train}} \subseteq R^{T_1 \times N}$ is segmented into a series of subsequences $X_{\text{train}} = \{X_{\text{train}}^i, i = 1, 2, \dots, m\} \subseteq R^{s_w \times N}$ by a sliding window, where X_{train}^i means $X_{t-s_w:t}$ when the window size is set to s_w . Given step length s_t , the number of subsequences can be calculated by $m = \lfloor (T_1 - s_w) / s_t \rfloor + 1$. Similarly, the testing dataset $\Phi_{\text{test}} \subseteq R^{T_2 \times N}$ is divided into subsequences $X_{\text{test}} = \{X_{\text{test}}^j, j = 1, 2, \dots, n\}$ by a sliding window, where $n = \lfloor (T_2 - s_w) / s_t \rfloor + 1$. For validation purposes, each point in the test dataset is marked with a binary number (0 for normal and 1 for abnormal).

B. Variable Correlation Analysis

Due to the inconsistent periodic nature of telemetry variables, the data detection window cannot cover the entire period. It is necessary to perform correlation analysis on the original training data before sliding the data window. The MIC [13] is an effective method for measuring the correlation between two variables.

For a given training dataset $\Phi_{\text{train}} \subseteq R^{T_1 \times N}$, for any two discrete variables with T_1 elements, the MIC values between $A = \{a_i \mid i = 1, \dots, T_1\}$ and $B = \{b_i \mid i = 1, \dots, T_1\}$ can be obtained by the following equation:

$$MI(A, B) = \sum_{a \in A} \sum_{b \in B} p(a, b) \log \frac{p(a, b)}{p(a)p(b)} \quad (2)$$

where $p(a, b)$ is the joint probability density of variables a and b , and $p(a)$ and $p(b)$ are the marginal probability densities of variables a and b , respectively, computed by the histogram estimation method.

For a finite set $D = \{(a_i, b_i), i = 1, \dots, n\}$, given a grid G , we can partition the a_i values of D into the a bin and the b_i values of D into the b bin. The MIC is given by

$$MI^*(D, x, y) = \max MI(D \mid G) \quad (3)$$

$$MIC(D) = \max_{xy < B(T_1)} \{M(D)_{x,y}\}. \quad (4)$$

The maximum information coefficient is the highest normalized MI value obtained in the feature matrix. where $\omega(1) < B(T_1) < O(T_1^{1-\varepsilon})$ and $0 < \varepsilon < 1$. In general, MIC works well in practice when $B(T_1) = T_1^{0.6}$. After the calculation, the MIC values between N telemetry variables and the correlation matrix M can be obtained. m_{ij} denotes the correlation between telemetry variable i and telemetry variable j , and it has a value between 0 and 1.

C. MAG Model

During the training phase, each telemetry variable within a time window exhibits both temporal and intrinsic property features. Representing the intrinsic properties of each telemetry variable using embedding vectors enables subsequent feedback, performance differentiation, and updates. Moreover, the attention mechanism can be employed to express the relationships between variables within the window more effectively. Therefore, we introduce an embedding vector for each telemetry variable to represent its features $v_i \in \mathbb{R}^d$, for $i \in \{1, 2, \dots, N\}$. These embeddings are randomly initialized and then trained together with the rest of the model. The similarity between these embeddings v_i represents the intrinsic properties of the telemetry variables.

In the MAG model, for a subseries combination of X_{train} or X_{test} , define the model input as the historical subseries data $\mathbf{x}^{(t)} := [x^{(t-w)}, x^{(t-w+1)}, \dots, x^{(t-1)}]$ for a sliding window of size w at time t . The calculation of the attention coefficient $\alpha_{i,j}$ is as follows:

$$\pi(i, j) = \text{ReLU} \left((v_i \oplus Wx_i^{(t)}) \oplus (v_j \oplus Wx_j^{(t)}) \right) \quad (5)$$

$$\alpha_{i,j} = \frac{\exp(\pi(i, j))}{\sum_{k \in \mathcal{N}(i) \cup \{i\}} \exp(\pi(i, k))}. \quad (6)$$

After computing the attention coefficient $\alpha_{i,j}$, it is combined with the correlation coefficient m_{ij} to form each edge e_{ij} and construct adjacency matrix E

$$e_{ij} = \alpha_{i,j} \cdot m_{ij}. \quad (7)$$

Following the construction of the edges, we employ LSTM technique to extract temporal association features. $\mathbf{x}^{(t)}$ is subjected to LSTM network to extract the temporal features. Temporal features can be extracted through the LSTM network denoted as

$$\mathbf{y}^{(t)} := [y^{(t-w)}, y^{(t-w+1)}, \dots, y^{(t-1)}]. \quad (8)$$

Finally, a graph neural network is employed to integrate the temporal and spatial associations within the subsequence. The constructed graph model utilizes the previously mentioned $e_{i,j}$ to form the adjacency matrix. The temporal feature $\mathbf{y}^{(t)}$ is used as input for the graph neural network, enabling the aggregation and update of information from each node and its neighbors, ultimately producing the output of node i represented $\mathbf{z}_i^{(t)}$ as follows:

$$\mathbf{z}_i^{(t)} = \text{ReLU} \left(e_{i,i} \mathbf{W}_i^{(t)} \mathbf{y}_i^{(t)} + \sum_{j \in \mathcal{N}(i)} e_{i,j} \mathbf{W}_j^{(t)} \mathbf{y}_j^{(t)} \right) \quad (9)$$

where $\mathbf{y}_i^{(t)} \in \mathbb{R}^w$ is the temporal input feature of node i , $\mathcal{N}(i) = \{j \mid e_{ij} > 0\}$ is the set of neighbors to node i , and $\mathbf{W} \in \mathbb{R}^{d \times w}$ is the trainable weight matrix of the shared linear transformation for each node.

From the above feature extractor, we obtain a representation of all N nodes as $\{\mathbf{z}_1^{(t)}, \dots, \mathbf{z}_N^{(t)}\}$. For each $\mathbf{z}_i^{(t)}$, we multiply it by the proceeding elements of the corresponding time series of embedding v_i (denoted as $\circ$). Then, we use the results across all nodes as input to a stacked fully connected layer with output dimension N to predict the vector of telemetry values at time step t

$$\hat{\mathbf{x}}^{(t)} = f_\theta \left([v_1 \circ \mathbf{z}_1^{(t)}, \dots, v_N \circ \mathbf{z}_N^{(t)}] \right). \quad (10)$$

The predicted output of the model is denoted as $\hat{\mathbf{x}}^{(t)}$. We denote the analog variables and state variables described in Section II as $x_a^{(t)}$ and $x_s^{(t)}$, respectively. Considering the heterogeneity between these two types, different loss functions are adopted. The analog variables utilize mean square error, minimizing the discrepancy between the predicted output $\hat{x}_a^{(t)}$ and the observed data $x_a^{(t)}$. On the other hand, for state variables, binary cross-entropy loss is used. In order to mitigate overfitting and to ensure appropriate sizing of the final edges within the sliding window, we introduce a constraint term. This term takes into account the acyclicity of the corresponding graph [14], and incorporates the findings of correlation analysis. The final loss function is as follows:

$$L_{\text{MSE}} = \frac{1}{N_a} \sum_{i=1}^{N_a} \left\| \hat{x}_{ai}^{(t)} - x_{ai}^{(t)} \right\|_2^2 \quad (11)$$

$$L_{\text{BCE}} = -\frac{1}{N_s} \sum_{i=1}^{N_s} \left[x_{si}^{(t)} \log \hat{x}_{si}^{(t)} + (1 - x_{si}^{(t)}) \log (1 - \hat{x}_{si}^{(t)}) \right] \quad (12)$$

$$h(E) = \text{tr}(e^{E \circ E}) - \text{tr}(e^{M \circ M}) \quad (13)$$

$$L_{\text{total}} = L_{\text{MSE}} + L_{\text{BCE}} + \lambda h(E) + \frac{c}{2} |h(E)|^2 \quad (14)$$

where λ and c represent Lagrangian multipliers and penalty parameters, and are solved by augmented Lagrangian method [15]. N_s represents the dimensionality of the state variables, and N_a represents the dimensionality of the analog variables. The network structure of the MAG is shown in Fig. 3.

D. Abnormal Scores

After the graph structure is acquired by the network, detection of the anomalies that deviate from normal patterns is the next step. The model achieves this by computing separate outliers for each telemetry and then amalgamating them into a solitary outlier per timestamp. Since our detection algorithm chooses an MAG-based model, it is also necessary to design outlier determination rules specifically based on the characteristics of that model. The anomaly score compares the expected behavior at moment t with the observed behavior and calculates the error

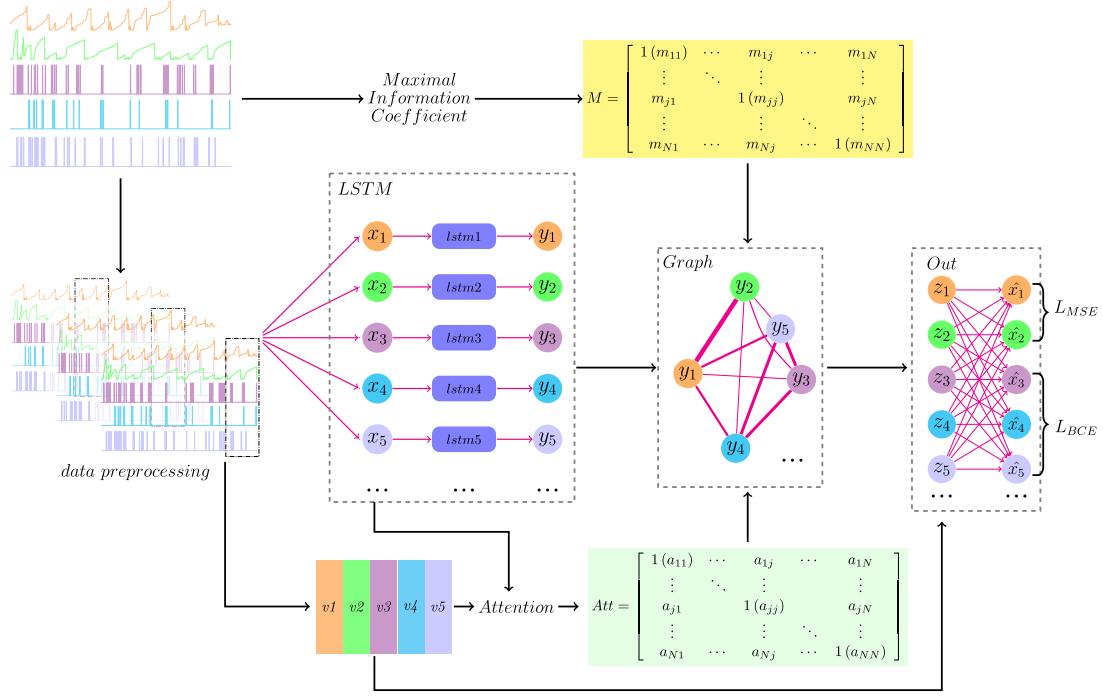


Fig. 3. Overall architecture of the proposed MAG anomaly detection algorithm.

value $Err(t)$ between moment t and the true telemetry value

$$Err(t) = \frac{1}{N_s} \sum_{i=1}^{N_s} |x_{si}^{(t)} - \hat{x}_{si}^{(t)}| + \frac{1}{N_a} \sum_{i=1}^{N_a} |x_{ai}^{(t)} - \hat{x}_{ai}^{(t)}|. \quad (15)$$

In order to prevent any one telemetry value from producing a deviation too high above the other telemetry values, we normalize the error value $Err(t)$ for each telemetry value to obtain $a(t)$.

In the threshold selection part, to avoid introducing additional hyperparameters, we propose in our experiments a method that is more computationally convenient. By calculating the deviation $a_{\text{train}}(t)$ of each timestamp on the training set, we can calculate the threshold value by the following equation to obtain:

$$\tau = \max \left(c_v^{\frac{1}{2}} (a_{\text{train}}(t) - \tilde{\mu}) \right) \quad (16)$$

$$c_v = \frac{\tilde{\mu}}{\tilde{\sigma}} \quad (17)$$

where c_v is the coefficient of variation, which is a statistical measure of the degree of variation in the telemetry data. $\tilde{\mu}$ and $\tilde{\sigma}$ are the median and interquartile range (IQR) of the values $a_{\text{train}}(t)$, respectively. We use the median and IQR instead of the mean and standard deviation needed in the coefficient of variation because they do not assume data distribution and are more robust to the exception scores of the model. Finally, if the $Err(t)$ of the test set exceeds any fixed threshold τ , then time t is marked as an exception.

Many published anomaly detection algorithms [16] use the 3σ algorithm and the peak over threshold (POT) algorithm to mine the threshold τ . POT is a threshold mining method using extreme

value theory, which assumes that the peaks in the time series satisfy the generalized Patoli distribution (GPD). Nonetheless, the suitability of the POT method may be limited when the distribution characteristics of the data are at variance with the GPD. For telemetry data of spacecraft systems, the experiments in the next section show that our approach is more adaptable.

The process of the MAG-based model algorithm is shown in Algorithm 1.

V. EXPERIMENTS

A. Datasets and Performance Index

We conducted experiments on two new telemetry datasets and two public datasets. The two new datasets, SCC-1 and SCC-2, come from the telemetry data of two different satellite systems. The two public telemetry datasets are provided by NASA for the Soil Moisture Active Passive (SMAP) satellite and the Mars Science Laboratory (MSL) rover [17]. The details of four datasets are shown in Table I. For these datasets, normal data points are marked as 0 and outliers are marked as 1. Then, 10% of the training dataset is divided for the validation dataset. Note that the training dataset contains only normal time series. We use commonly used metrics to evaluate the performance of our proposed MAG algorithm, which are Precision, Recall, and F1 score

$$\text{Precision} = \frac{TP}{TP + FP} \quad (18)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (19)$$

Algorithm 1: Calculate MAG-Based Model.**Variable Correlation Analysis:**

The training dataset Φ_{train} is calculated using MIC analysis to obtain the correlation coefficient matrix M .

Train Stage:

for l^{th} epoch in train epochs **do**

1. The attention mechanism applied to the fused the subsequence X_{train} and node embedding vector V_{train} to obtain attention coefficient matrix Att , which is combined with M to get edges E .
2. Compute the temporal feature Y_{train} of X_{train} .
3. Y_{train} and E are fed into the graph neural network for message passing, node aggregation, and node update to calculate output Z_{train} .
4. Multiply Z_{train} and V_{train} point by point to get $\hat{X}_{train}$ and calculate L_{MSE} and L_{BCE} to get L_{total} .
5. Update the parameters in the neural network by L_{total} .

end for

Threshold Calculation:

for X_{train}^i in X_{train} **do**

Utilize the real value $x^{(t)}$ and the trained network to compute the predicted value $\hat{x}^{(t)}$ and the error $Err(t)$.

end for

Normalize $Err(t)$ to get $a(t)$ and calculate the anomaly threshold τ .

Testing Stage:

for X_{test}^i in X_{test} **do**

1. Utilize the real value $x^{(t)}$ and the trained network to compute the predicted value $\hat{x}^{(t)}$ and the error $Err(t)$.
2. The outlier value is obtained by making a difference comparison with the threshold τ .

end for

TABLE I
DETAILS OF DATA

Dataset	Variable number	Analog number	State number	Train dataset	Test dataset	Anomaly Ratio(%)
SCC-1	57	8	49	400000	500000	0.125
SCC-2	27	1	26	26432	57853	0.108
SMAP	25	1	24	135183	427567	0.128
MSL	55	1	54	58267	73679	0.105

$$F1 = \frac{2 \cdot \text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} \quad (20)$$

where TP is the number of predicted actual anomalies, FP is the number of the false positive samples, and FN is the number of the false negative samples.

B. Experimental Settings and Platform

In the data preprocessing stage, we set the window size $s_w = 50$ and the step size $s_t = 1$, and then split the original time series into the desired subsequences. In the network structure, we set the embedding vector dimension to 128 and set the hidden layer to 128. We are utilizing the Adam optimizer [18] with a learning rate of 1×10^{-3} to train the model. We use a widely used

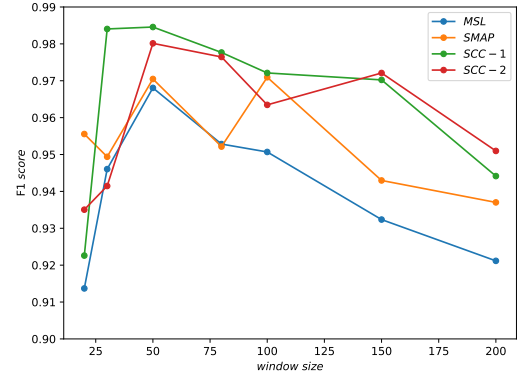


Fig. 4. Relationship between F1 scores and window length for the four datasets.

tuning strategy [19], [20]: if a certain time point in a continuous anomaly segment is detected, all anomalies in that segment are considered to be correctly detected. Based on the observation that the time point of the anomaly will raise an alert and further make the whole segment noticed in realistic applications, this strategy is reasonable. We implemented our approach and its variants in PyTorch [21] version 1.9.1 using CUDA 11.6 and PyTorch Geometric Library [22]. We set the training model to 100 cycles and used 10 as an early stop. To obtain reliable results and reduce the randomness of the training phase, the samples were trained and tested ten times separately, and then the standard deviation of the performance metrics was calculated. Finally, our algorithm is trained and tested on a server with Intel(R) Xeon(R) CPU E5-2690 v4 @ 2.60 GHz and NVIDIA RTX 3090 graphics card.

C. Results and Comparisons

1) **Window Size:** To determine the appropriate window size S_w , we conducted experiments on three datasets by selecting a portion of small datasets. The window sizes were set to 20, 30, 50, 80, 100, 150, and 200.

The F1 scores for the four datasets are depicted in Fig. 4. Opting for an excessively long window can result in redundant information, slow response, and increased computational complexity. Conversely, selecting a window length that is too short will result in an inadequate capture of temporal features, leading to insufficient stability. The experimental results from the four small datasets imply that a window length of 50 is optimal.

2) **Baseline Comparison:** To exhibit the effectiveness of our proposed algorithm, we conduct a comparative analysis of its performance with other benchmark unsupervised anomaly detection algorithms. These include deep learning based models such as AnomalyTransformer [19], ST-GAN [9], InterFusion [23], GDN [11], and GRU-VAE [8]; clustering-based techniques such as Deep-SVDD [24]; as well as classical methods such as OC-SVM [25] and IsolationForest [26]. Notably, ST-GAN and AnomalyTransformer represent the most sophisticated deep models. We evaluate the average Precision, Recall, and F1-score after ten rounds on the SCC-1, SCC-2, SMAP, and

TABLE II
QUANTITATIVE RESULTS OF MAG (OURS) ACROSS FOUR REAL TELEMETRY DATASETS. HEREIN, P, R, AND F1 SIGNIFY THE PRECISION, RECALL, AND F1-SCORE, RESPECTIVELY

Model	MSL			SMAP			SCC-1			SCC-2		
	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
OC-SVM	58.25	56.82	69.72	53.48	59.81	56.46	60.58	66.27	63.29	70.26	55.23	61.84
IsolationForest	54.23	85.46	66.35	53.25	59.86	56.36	73.54	64.38	68.65	69.85	58.30	63.55
Deep-SVDD	92.23	77.85	84.43	89.21	55.42	68.36	87.25	86.34	86.79	82.31	73.49	77.65
GRU-VAE	86.35	80.21	83.16	93.20	69.93	79.90	89.28	81.59	85.26	84.53	79.39	81.87
GDN	89.23	79.15	83.88	84.34	76.72	80.34	90.48	72.23	80.33	88.24	73.31	80.08
InterFusion	87.32	92.50	89.83	89.77	88.52	89.14	91.13	84.35	87.60	89.23	88.42	88.82
ST-GAN	87.78	99.36	93.21	89.59	99.58	94.32	85.18	86.17	85.67	90.12	94.23	92.12
Transformer	92.09	95.15	93.59	94.13	99.40	96.69	93.58	96.71	94.85	93.23	97.12	95.13
Ours	95.94	97.98	96.95	95.26	99.98	97.56	97.35	99.72	98.52	97.67	98.87	98.27

A higher value in these three metrics indicates enhanced performance.

The bolded values represent the best values obtained for Precision, Recall, and F1-score.

TABLE III
RESULTS OF ABLATION AND COMPARISON EXPERIMENTS

Ablation	Method	MSL			SMAP			SCC-1			SCC-2		
		P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)	P(%)	R(%)	F1(%)
Edge mechanism	Attention Association	84.21	96.97	90.14	91.52	94.51	92.99	92.01	95.66	93.81	94.87	93.34	94.12
		95.58	97.60	96.58	94.60	99.98	97.21	97.14	99.32	98.21	97.31	98.73	98.01
Correlation analysis	Pearson Spearman	92.96	87.03	89.89	84.31	95.23	89.43	85.96	89.14	87.92	88.91	84.31	86.54
		94.21	87.23	90.58	87.35	87.51	87.42	89.76	88.95	89.35	92.31	86.21	89.15
Temporal characteristics analysis	NULL	65.41	62.53	63.93	59.76	63.23	61.44	62.32	67.31	64.71	69.78	71.24	70.50
	Linear	68.31	74.92	71.46	73.60	70.90	72.22	67.47	76.39	71.65	72.05	75.51	73.74
	RNN	90.88	89.03	89.95	89.67	95.08	92.29	94.88	89.44	92.08	90.85	94.15	92.47
	GRU	95.01	96.62	95.81	92.90	97.06	94.94	96.28	94.38	95.32	94.59	95.57	95.08
Loss function	MSE	91.88	94.14	93.00	93.59	97.27	95.40	93.08	89.36	91.18	96.64	88.35	92.31
	MAE	85.35	90.13	87.68	86.95	90.59	88.74	85.08	94.88	89.71	87.35	92.27	89.74
Ours		95.94	97.98	96.95	95.26	99.98	97.56	97.35	99.72	98.52	97.67	98.87	98.27

The bolded values represent the best values obtained for Precision, Recall, and F1-score.

MSL datasets. The comparative results of the four algorithms are showcased in Table II.

According to Table II, our methodology attains the highest F1 scores across all four datasets, demonstrating optimal performance in balancing false alarm and missed alarm rates. Moreover, our approach exhibits the most satisfactory results in recall rate on the SCC-1 and SMAP datasets, indicating a minimal likelihood of producing false alarms on these two datasets.

3) Ablation Comparison: To investigate the necessity of each component of our method, we gradually excluded and substituted these components and monitored how the model performance degraded. The experimental results are shown in Table III.

To compare the effectiveness of the edge mechanism, we conducted separate experiments by eliminating either the correlation analysis or attention mechanism. The experiment showed that the fusion of both mechanisms yielded the most favorable outcome, as the introduction of the attention mechanism after the correlation analysis effectively enhances the model's adaptability. However, solely relying on the attention mechanism and the data within the window is inadequate for effectively capturing the extent of correlation and dependency over prolonged periods.

Furthermore, we used the Pearson and Spearman methods [27] for comparing correlations. The experimental results indicate that the MIC outperforms the Pearson and Spearman correlation coefficients. This discrepancy arises because the Pearson and Spearman correlation coefficients assume linear correlation, causing inaccuracies when analyzing nonlinear relationships. On the contrary, MIC does not rely on such assumptions and can accurately capture both linear and nonlinear correlations over extended periods.

Regarding the temporal relationship analysis, we compared linear, RNN, and GRU layer connections [28], using ablation experiments. The findings indicate that employing the network layer or a linear layer results in less effectiveness, primarily due to the inability to extract temporal features. In contrast, incorporating layer connections of RNN, GRU, or LSTM yields more effective findings. Specifically, LSTM, with a more intricate and robust structure than GRU and RNN, provides better control over the flow of information and enables the capture of long-term dependencies. Therefore, LSTM can extract superior temporal features, resulting in better outcomes.

As for the loss function, we applied the mean squared error (MSE) and mean absolute error (MAE) [29] and compared

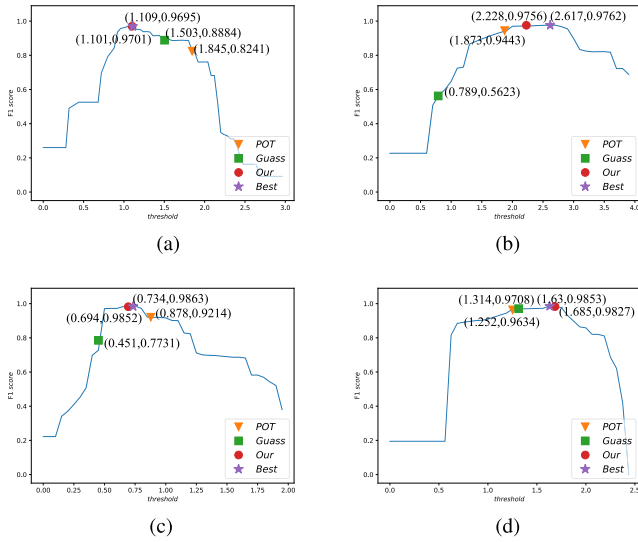


Fig. 5. Illustrative relationship between threshold and F1 score. (a) MSL dataset. (b) SMAP dataset. (c) SCC-1 dataset. (d) SCC-2 dataset.

them with our method. Experimental findings demonstrate that employing two mixed loss functions for two different types of data can significantly enhance the model's effectiveness.

4) Threshold Comparison: We also evaluated the impact of the threshold value τ on the F1-score of the four datasets. As shown in Fig. 5, when the threshold is too small, the Recall value will be low, resulting in a decreased F1-score. While if the threshold is set too high, the Precision value will decrease, causing a drop in F1-score. Therefore, it is crucial to set the threshold appropriately to ensure the optimal F1-score.

We conducted a comparison between two thresholding methods: 3σ thresholding based on the Gaussian distribution and extreme value theory thresholding (EVT). The impact of these thresholds on the F1 scores of the four test datasets is illustrated in Fig. 5. The results reveal that, as the threshold (τ) increases, the F1 score initially reaches its peak and then declines. Our proposed thresholding method closely approximates the optimal threshold acquired through exhaustive search, thus demonstrating the effectiveness of our anomaly threshold determination rule. This is likely because the 3σ thresholding and EVT methods both assume prior data distributions which are not applicable to telemetry data. Conversely, our method, which employs formulas utilizing the median and interquartile range, is more robust and unaffected by the shape of the data distribution.

VI. CONCLUSION

This article proposed a telemetry data anomaly detection algorithm based on the MAG structural model. Specifically, the algorithm constructed a graph-structured model that used embedding vectors to describe the intrinsic properties of each dimension, correlation analysis to investigate long-term dependencies, an attention mechanism to determine short-term interactions among dimensions, and LSTM to extract temporal

features. Finally, by fusing these modules through a graph neural network, the model effectively integrated the coupling relationships between the dimensions and temporal characteristics of these sequences, allowing for the successful detection of various types of anomalies. To determine anomalies, an anomaly score adapted to the network structure was introduced.

To ascertain the effectiveness and superiority of our proposed anomaly detection algorithm, we conducted experiments on four genuine telemetry datasets and compared our approach with other state-of-the-art algorithms, achieving the optimal results. Moreover, the ablation experiments provided further evidence of the effectiveness of our model components. Our proposed anomaly thresholds showcase superior accuracy and closely approximate the optimum thresholds when compared to other widely used techniques.

However, although it does not affect the efficiency of real-time anomaly detection, the computational cost increased with the increase in data volume during the computation of correlation analysis, warranting optimization of the computational efficiency of MIC. Additionally, investigating more MAG-based anomaly detection algorithms for various anomaly types and advancing into fault diagnosis presented promising research topics.

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